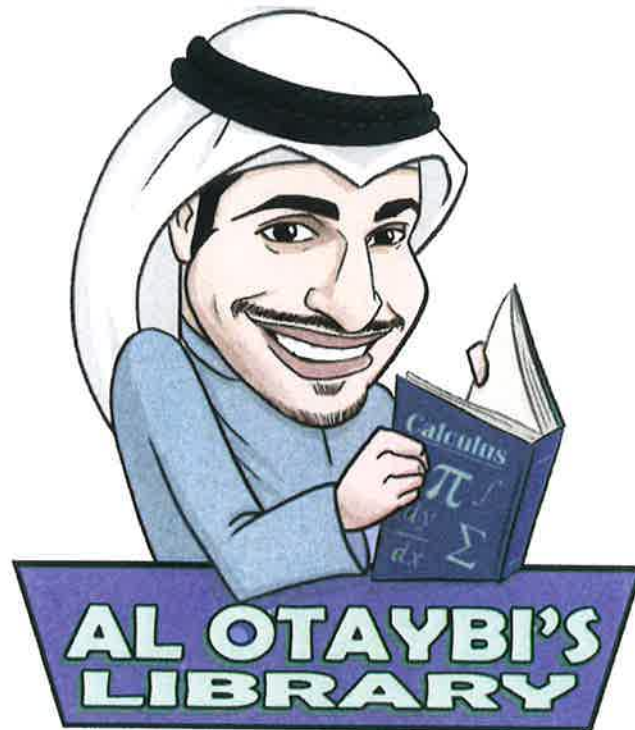


Kuwait University

2021/2022

Fall course



Calculus A

First

Sections : **1.2** (part b) - **1.4**

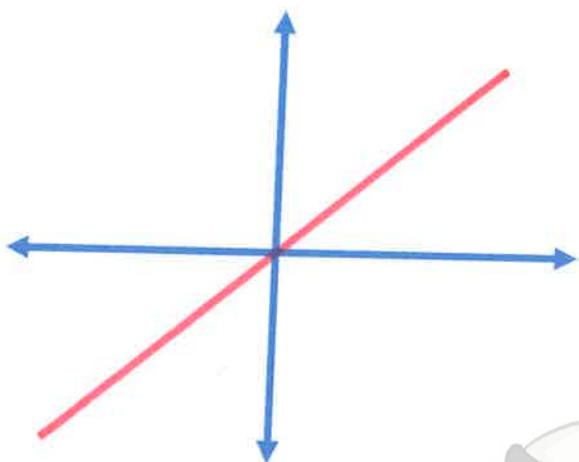
1.2 Mathematical Models: A Catalog of Essential Functions

1.4 Exponential Functions

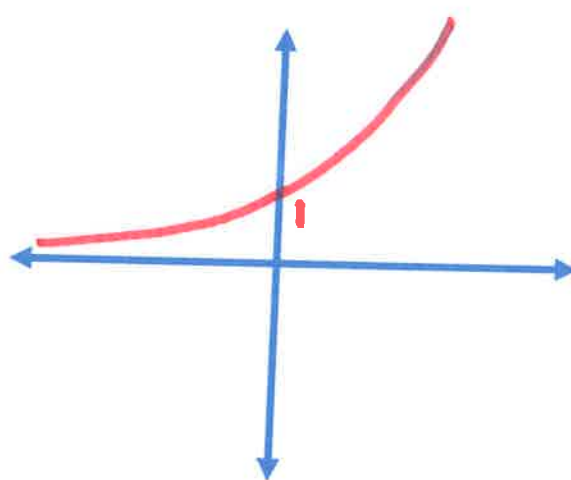
(2)

Graphs

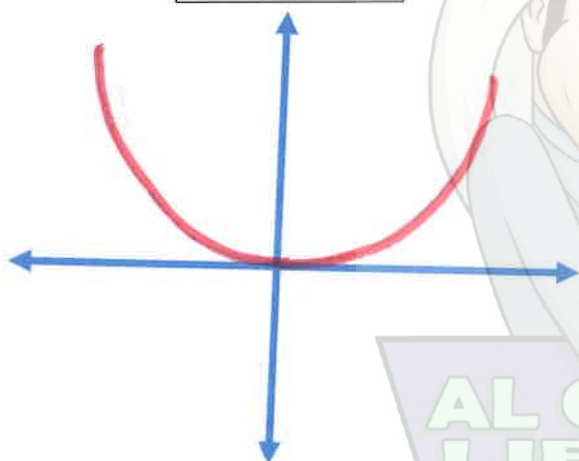
$$y = x$$



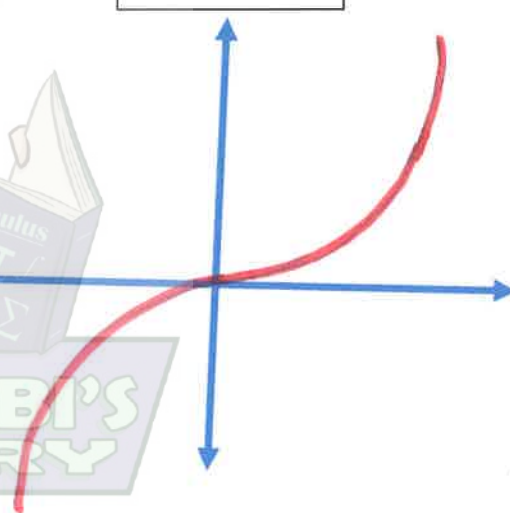
$$y = 5^x$$



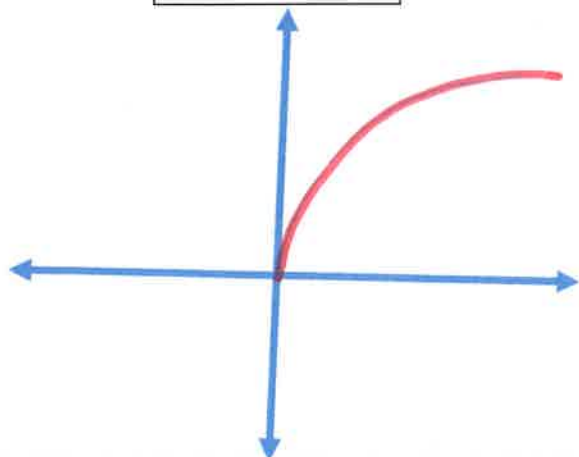
$$y = x^2 \text{ زوجي}$$



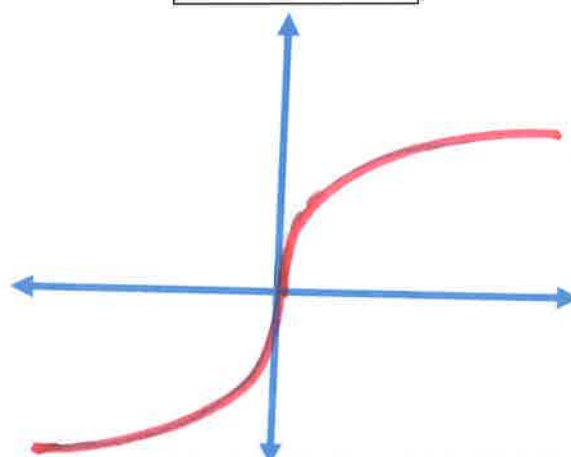
$$y = x^3 \text{ فردي}$$



$$y = \sqrt{x} \text{ زوجي}$$



$$y = \sqrt[3]{x} \text{ فردي}$$



AL OTAYBI'S
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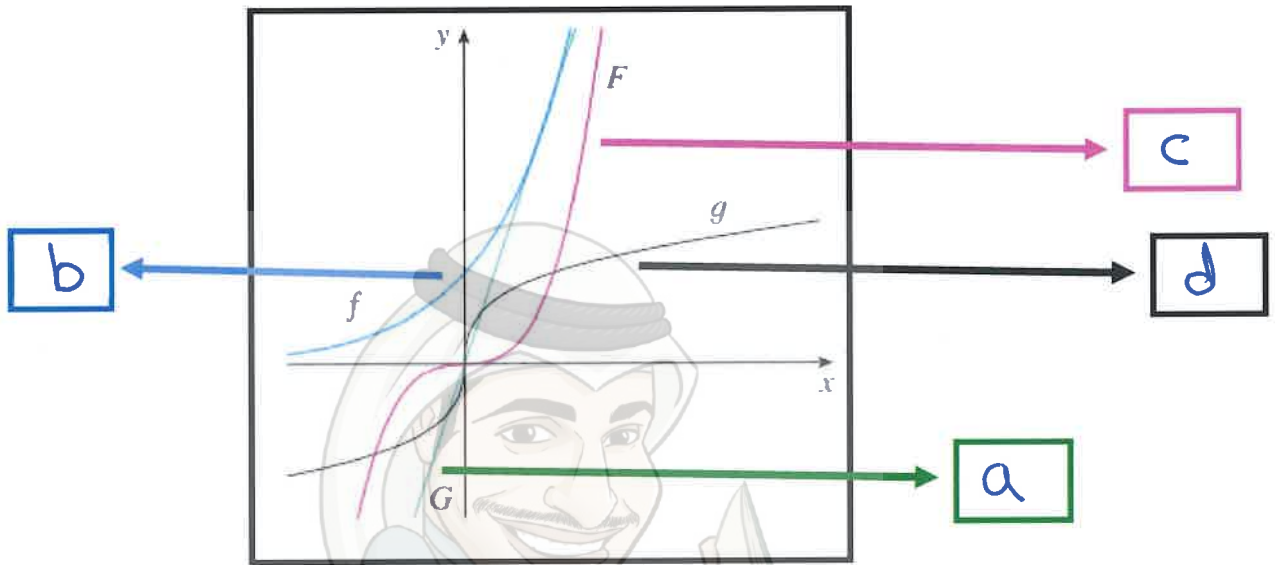


Text book questions



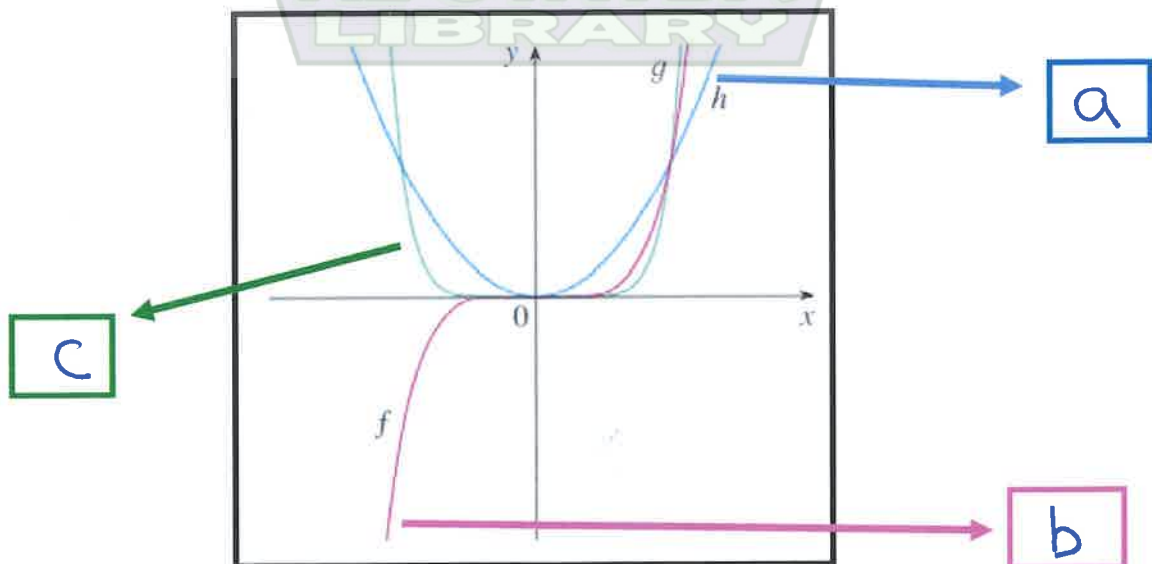
Match each equation with its graph.

a) $y = 3x$ b) $y = 3^x$ c) $y = x^3$ d) $y = \sqrt[3]{x}$



Match each equation with its graph.

a) $y = x^2$ b) $y = x^5$ c) $y = x^8$



Function	Domain
• حدوديات • مطلق • e^x a^x • $\tan^{-1}x$	$\mathbb{R}$
$\sqrt{x}$ فردي	$\mathbb{R}$
$\sqrt{x}$ زوجي	$x \geq 0$
• $\ln x$ • $\log_a x$	$x > 0$
• $\sin^{-1}x$ • $\cos^{-1}x$	$-1 \leq x \leq 1$
$\frac{\text{بسط}}{\text{مقام}}$	$\mathbb{R} / \{\text{أصفار المقام}\}$



ملاحظة مهمة



عندما وجود أكثر من دالة في وقت واحد (دالة مركبة) فإنه يتم
ايجاد مجال كل دالة بشكل منفصل عن الأخرى ثم يتم ايجاد تقاطعهم

Find the domain of

$$f(x) = \sin(x + 5)$$

$$D_f = \mathbb{R}$$

$$f(x) = \sqrt{x + 3}$$

$$x + 3 \geq 0$$

$$x \geq -3$$

$$\Rightarrow D_f = [-3, \infty)$$

$$f(x) = \frac{1}{x - 5}$$

$$x - 5 = 0$$

$$x = 5$$

$$\Rightarrow D_f = \mathbb{R} \setminus \{5\}$$

$$f(x) = \frac{x}{x + 7}$$

$$x + 7 = 0$$

$$x = -7$$

$$\Rightarrow D_f = \mathbb{R} \setminus \{-7\}$$

$$f(x) = \frac{x + 3}{x^2 - 4}$$

$$x^2 - 4 = 0$$

$$(x - 2)(x + 2) = 0$$

$$x = 2 \quad x = -2$$

$$\Rightarrow D_f = \mathbb{R} \setminus \{\pm 2\}$$

$$f(x) = \frac{3x + 1}{x^2 - x - 2}$$

$$x^2 - x - 2 = 0$$

$$(x + 1)(x - 2) = 0$$

$$x = -1 \quad x = 2$$

$$\Rightarrow D_f = \mathbb{R} \setminus \{-1, 2\}$$

Find the domain of each function

$$f(x) = \sqrt{x - 5}$$

$$x - 5 \geq 0$$

$$x \geq 5$$

$$D_f = [5, \infty)$$

$$f(x) = \sqrt{5 - x}$$

$$5 - x \geq 0$$

$$5 \geq x$$

$$D_f = (-\infty, 5]$$

$$f(x) = \ln(x - 3)$$

$$x - 3 > 0$$

$$x > 3$$

$$D_f = (3, \infty)$$

$$f(x) = \log_3(3 - x)$$

$$3 - x > 0$$

$$3 > x$$

$$D_f = (-\infty, 3)$$

$$f(x) = \frac{x + 3}{x - 3}$$

$$x - 3 = 0$$

$$x = 3$$

$$D_f = \mathbb{R} \setminus \{3\}$$

$$f(x) = \frac{x + 2}{x^2 - 4}$$

$$x^2 - 4 = 0$$

$$x^2 = 4 \rightarrow x = \pm 2$$

$$D_f = \mathbb{R} \setminus \{\pm 2\}$$

$$f(x) = \sin^{-1}(x - 2)$$

$$-1 \leq x - 2 \leq 1$$

$$1 \leq x \leq 3$$

$$D_f = [1, 3]$$

$$f(x) = \cos^{-1}(x - 1)$$

$$-1 \leq x - 1 \leq 1$$

$$0 \leq x \leq 2$$

$$D_f = [0, 2]$$

Find the domain of each function

$$f(x) = \frac{1}{e^x - 1}$$

$$e^x - 1 = 0$$

$$e^x = 1$$

$$e^x = e^0$$

$$x = 0 \rightarrow D_f = \mathbb{R} \setminus \{0\}$$

$$f(x) = \ln(5^x - 1)$$

$$5^x - 1 > 0$$

$$5^x > 1$$

$$5^x > 5^0$$

$$x > 0 \rightarrow D_f = (0, \infty)$$

$$f(x) = \frac{x+2}{x^2+4}$$

$$x^2 + 4 = 0$$

$$x^2 = -4$$

$$\therefore D_f = \mathbb{R}$$

$$f(x) = \sqrt{x^2 + 9}$$

$$x^2 + 9 \geq 0$$

$$x^2 \geq -9 \rightarrow D_f = \mathbb{R}$$

$$f(x) = \frac{x}{5^x - 1}$$

$$5^x - 1 = 0$$

$$5^x = 1$$

$$5^x = 5^0$$

$$x = 0 \rightarrow D_f = \mathbb{R} \setminus \{0\}$$

$$f(x) = \ln(e^x - 1)$$

$$e^x - 1 > 0$$

$$e^x > 1$$

$$e^x > e^0$$

$$x > 0 \rightarrow D_f = (0, \infty)$$

$$f(x) = \frac{2}{e^x + 4}$$

$$e^x + 4 = 0$$

$$e^x = -4$$

$$D_f = \mathbb{R}$$

$$f(x) = \ln(e^x + 3)$$

$$e^x + 3 > 0$$

$$e^x > -3 \rightarrow D_f = \mathbb{R}$$



Text book questions



Find the domain of each function.

$$f(x) = \frac{1 - e^{x^2}}{1 - e^{1-x^2}}$$

$$1 - e^{1-x^2} = 0$$

$$e^{1-x^2} = 1$$

$$e^{1-x^2} = e^0$$

$$1 - x^2 = 0$$

$$x^2 = 1$$

$$x = \pm 1$$

$$D_f = \mathbb{R} \setminus \{\pm 1\}$$

Find the domain of each function.

$$f(x) = \frac{1+x}{e^{\cos x}}$$

$$e^{\cos x} = 0 \quad \text{No value of } x$$

$$D_f = \mathbb{R}$$

Find the domain of each function.

$$g(t) = \sqrt{10^t - 100}$$

$$10^t - 100 \geq 0$$

$$10^t \geq 100$$

$$10^t \geq 10^2$$

$$t \geq 2$$

$$D_g = [2, \infty)$$

Find the domain of each function.

$$g(t) = \sin(e^t - 1)$$

$$D_g = \mathbb{R}$$

Find the domain of

$$f(x) = \sin^{-1}(x - 1)$$

$$\begin{aligned} -1 &\leq x-1 \leq 1 \\ 0 &\leq x \leq 2 \\ \Rightarrow D_f &= [0, 2] \end{aligned}$$

Find the domain of

$$f(x) = \cos^{-1}(2x + 1) - \tan^{-1}x$$

$$\begin{aligned} -1 &\leq 2x+1 \leq 1 \\ -2 &\leq 2x \leq 0 \\ \div 2 \quad -1 &\leq x \leq 0 \\ &[-1, 0] \end{aligned}$$

$\mathbb{R}$

$\Rightarrow D_f = [-1, 0]$

Tricky Question !!

Find the domain of

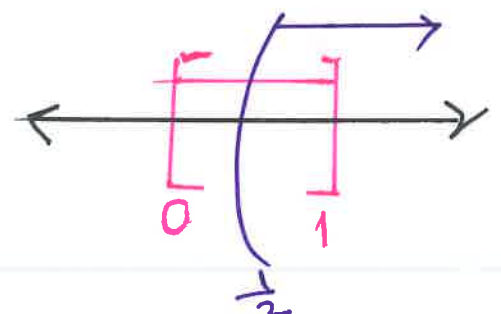
$$f(x) = \cos^{-1}(2x - 1) - \ln\left(x - \frac{1}{2}\right)$$

$$\begin{aligned} -1 &\leq 2x-1 \leq 1 \\ 0 &\leq 2x \leq 2 \\ \div 2 \quad 0 &\leq x \leq 1 \end{aligned}$$

$[0, 1]$

$$\begin{aligned} x - \frac{1}{2} &> 0 \\ x &> \frac{1}{2} \\ &(\frac{1}{2}, \infty) \end{aligned}$$

$$\Rightarrow D_f = (\frac{1}{2}, 1]$$



Find the domain of

$$h(x) = \sqrt{5-x} + \sqrt{x+4}$$

$$5-x \geq 0$$

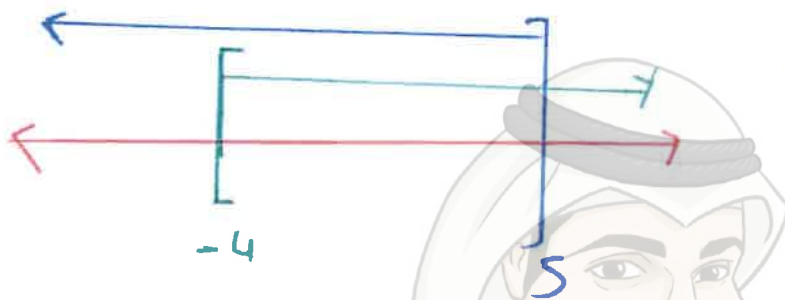
$$5 \geq x$$

$$(-\infty, 5]$$

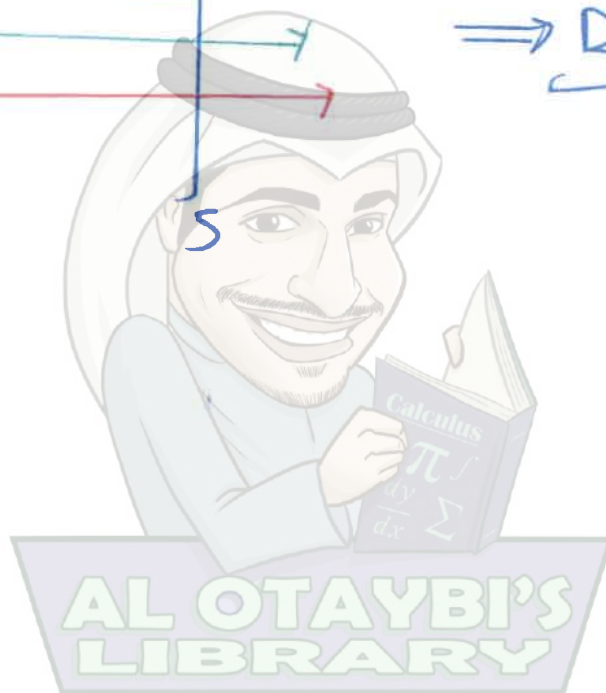
$$x+4 \geq 0$$

$$x \geq -4$$

$$[-4, \infty)$$



$$\Rightarrow D_f = [-4, 5]$$



Find the domain of

$$f(x) = \sin x^2 + \sqrt[3]{x+10} - \sqrt{x-1} + \frac{x+2}{x-5}$$

$\mathbb{R}$

$\mathbb{R}$

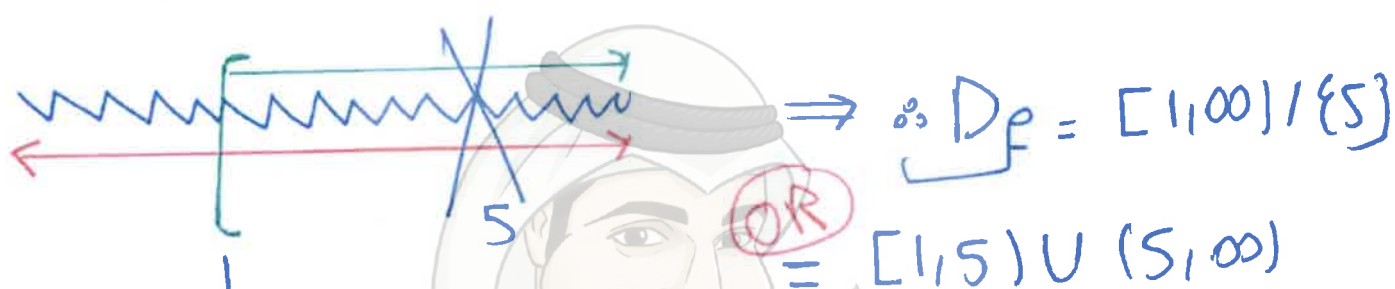
$$x-1 \geq 0$$

$$x \geq 1$$

$$x-5 \neq 0$$

$$x \neq 5$$

$$[1, \infty) \cap \mathbb{R} \setminus \{5\}$$



$$f(x) = x^2 - \frac{\sqrt{x-2}}{\sqrt{x-7}}$$

$\mathbb{R}$

$$x-2 \geq 0$$

$$x \geq 2$$

$$[2, \infty)$$

$$x-7 \geq 0$$

$$x \geq 7$$

$$[7, \infty)$$

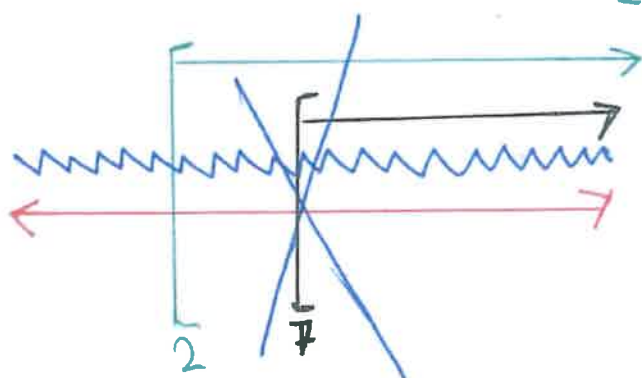
المقام

$$\sqrt{x-7} = 0$$

$$x-7=0$$

$$x=7$$

$$\mathbb{R} \setminus \{7\}$$





Laws of Exponent



$$(1) a^x a^y = a^{x+y}$$

$$(2) \frac{a^x}{a^y} = a^{x-y}$$

$$(3) (a^x)^y = a^{xy}$$

$$(4) (ab)^x = a^x b^x$$

$$(5) \left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$$

$$(6) a^{-x} = \frac{1}{a^x}$$



مصيبة .. كارثة .. جريمة !!



الأس لا يمكن توزيعه على
الجمع أو الطرح

$$(a \pm b)^x = a^x \pm b^x$$

Use the law of exponent to simplify the expression

$$\frac{x^{2n} x^{3n-1}}{x^{n+1}}$$

$$\begin{aligned}
 &= \frac{x^{2n+3n-1}}{x^{n+1}} = \frac{x^{5n-1}}{x^{n+1}} \\
 &= x^{5n-1-(n+1)} = x^{5n-1-n-1} \\
 &= x^{4n-2}
 \end{aligned}$$



Text book question



Use the Law of Exponents to rewrite and simplify the expression.

$$x(3x^2)^3$$

$$\begin{aligned}
 &= x \cdot 3^3 \cdot (x^2)^3 \\
 &= x \cdot (27) \cdot x^6 \\
 &= 27 x^7
 \end{aligned}$$

$$b^8(2b)^4$$

$$\begin{aligned}
 &= b^8 \cdot 2^4 \cdot b^4 \\
 &= 16 b^{12}
 \end{aligned}$$

Use the Law of Exponents to rewrite and simplify the expression.

$$\frac{\sqrt{a}\sqrt{b}}{\sqrt[3]{ab}}$$

$$= \frac{(a \cdot b^{\frac{1}{2}})^{\frac{1}{2}}}{(a \cdot b)^{\frac{1}{3}}}$$

$$= \frac{a^{\frac{1}{2}} (b^{\frac{1}{2}})^{\frac{1}{2}}}{a^{\frac{1}{3}} \cdot b^{\frac{1}{3}}}$$

$$= \frac{a^{\frac{1}{2}} b^{\frac{1}{4}}}{a^{\frac{1}{3}} b^{\frac{1}{3}}}$$

$$\begin{aligned}
 &= a^{\frac{1}{2} - \frac{1}{3}} \cdot b^{\frac{1}{4} - \frac{1}{3}} \\
 &\quad \frac{\frac{1}{2} - \frac{1}{3}}{\frac{3}{6} - \frac{2}{6}} = \frac{1}{6} \qquad \frac{\frac{1}{4} - \frac{1}{3}}{\frac{3}{12} - \frac{4}{12}} = \frac{-1}{12}
 \end{aligned}$$

$$= a^{\frac{1}{6}} \cdot b^{-\frac{1}{12}} = \frac{a^{\frac{1}{6}}}{b^{\frac{1}{12}}} = \frac{\sqrt[6]{a}}{\sqrt[12]{b}}$$

Use the law of exponent to simplify the expression

$$8^{\frac{4}{3}}$$

$$= (2^3)^{\frac{4}{3}}$$

$$= 2^4 = 16$$

$$\frac{4^{-3}}{2^{-8}}$$

$$= \frac{2^8}{4^3}$$

$$= \frac{2^8}{(2^2)^3}$$

$$= \frac{2^8}{2^6}$$

$$= 2^{8-6}$$

$$= 2^2 = 4$$

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LIBRARY



Text book question



Use the Law of Exponents to rewrite and simplify the expression.

$$\frac{(6y^3)^4}{2y^5}$$

$$= \frac{6^4 (y^3)^4}{2y^5}$$

$$= \frac{((3)(2))^4 y^{12}}{2y^5}$$

$$= \frac{3^4 \cancel{2^4} \cancel{y^{12}}}{\cancel{2} \cancel{y^5}}$$

$$= 3^4 2^3 y^7$$

$$= 81 (8) y^7$$

$$= 648 y^7$$

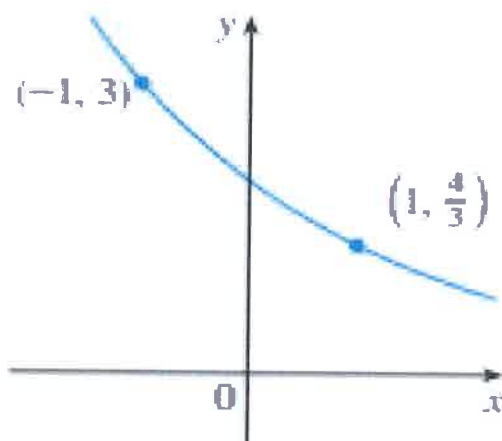
$$\begin{array}{r} 81 \\ 8 \\ \hline 648 \end{array}$$



Text book question



Find the exponential function $f(x) = Cb^x$ whose graph is given



$$y = Cb^x$$

at $(-1, 3)$

$$3 = Cb^{-1}$$

$$3 = \frac{C}{b} \rightarrow C = 3b \quad (1)$$

at $(1, \frac{4}{3})$

$$\frac{4}{3} = Cb^1 \quad (2)$$

(1) in (2)

$$\frac{4}{3} = 3bb$$

$$\frac{4}{3} = 3b^2$$

$$\times \frac{1}{3}$$

$$\frac{1}{3} \frac{4}{3} = \frac{1}{3} 3b^2$$

$$\frac{4}{9} = b^2$$

$$b = \pm \sqrt{\frac{4}{9}}$$

$$b = \pm \frac{2}{3}$$

$$b = \frac{2}{3}$$

$$b = -\frac{2}{3} \text{ rejected}$$

$$\therefore \boxed{b = \frac{2}{3}}$$

in (1)

$$C = 3b$$

$$C = 3 \frac{2}{3} \rightarrow \therefore \boxed{C = 2}$$

$$y = Cb^x$$

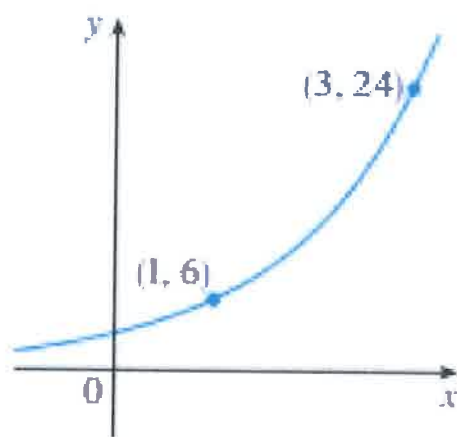
$$y = 2 \left(\frac{2}{3}\right)^x$$



Text book question



Find the exponential function $f(x) = Cb^x$ whose graph is given



$$y = Cb^x$$

at (1, 6)

$$6 = Cb^1 \rightarrow C = \frac{6}{b} \quad (1)$$

at (3, 24)

$$24 = Cb^3 \quad (2)$$

(1) in (2)

$$24 = \frac{6}{b} b^3$$

$$\times \frac{1}{6} \quad 24 = 6b^2$$

$$4 = b^2$$

$$b = \pm 2$$

$$b = 2$$

$$b = -2$$

rejected

$$b = 2$$

in (1)

$$C = \frac{6}{b}$$

$$C = \frac{6}{2} \rightarrow C = 3$$

$$C = 3$$

$$y = Cb^x$$

$$y = 3(2)^x$$



Text book question



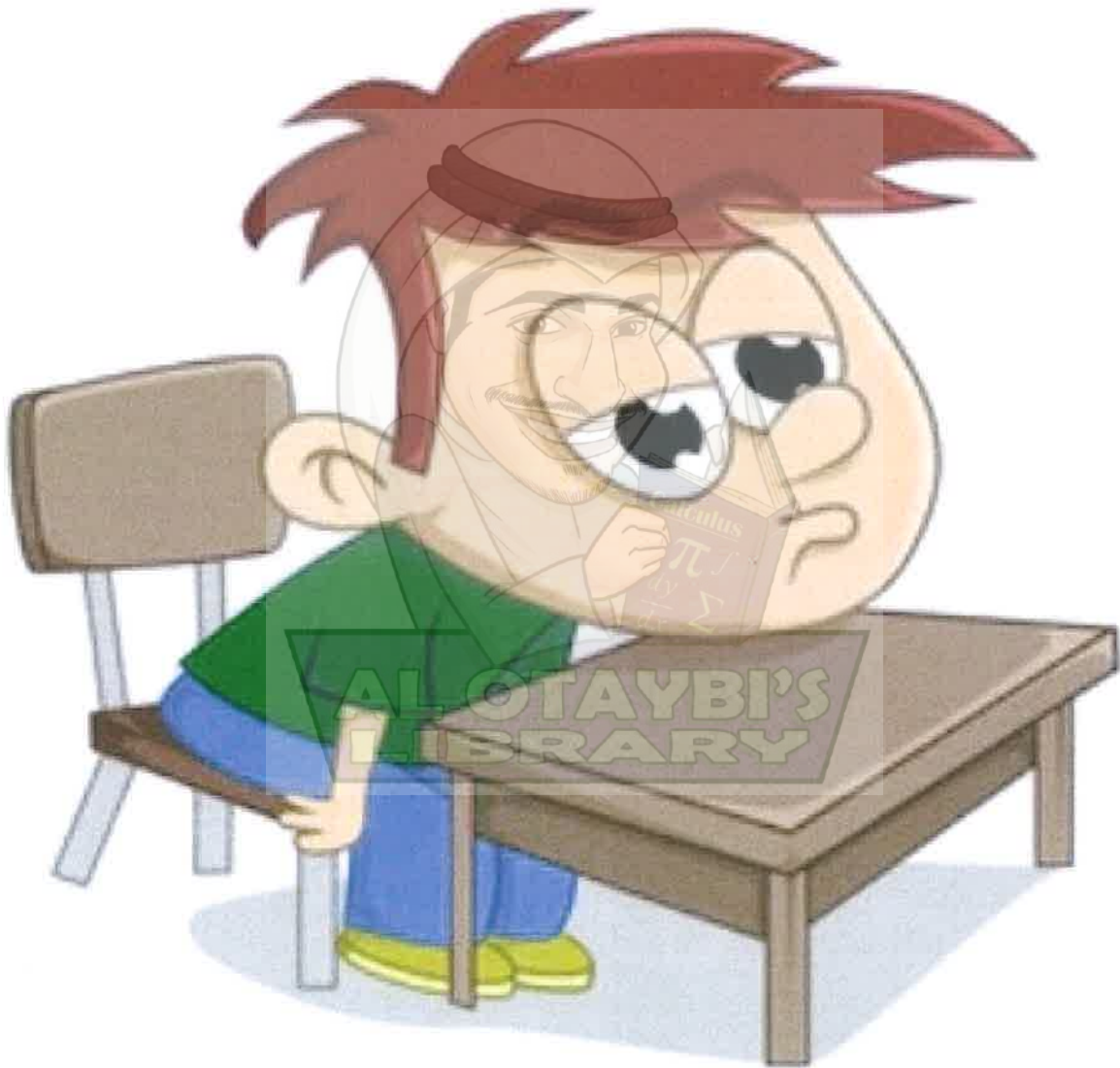
If $f(x) = 5^x$, Show that

$$\frac{f(x+h) - f(x)}{h} = 5^x \left(\frac{5^h - 1}{h} \right)$$

$$\begin{aligned} \text{L.H.S} &= \frac{f(x+h) - f(x)}{h} \\ &= \frac{5^{x+h} - 5^x}{h} \\ &= \frac{5^x \cdot 5^h - 5^x}{h} \\ &= \frac{5^x (5^h - 1)}{h} \\ &= 5^x \left(\frac{5^h - 1}{h} \right) = \text{R.H.S} \end{aligned}$$

تكملة سكرشن

1.2





Zoom on
Linear function
(Polynomial of degree 1)



To find the x-intersection and y-intersection



أعوض مكان الـ y بصفر

أعوض مكان الـ x بصفر

Example



Find the x intercept and the y intercept of

$$f(x) = 5x - 10$$

$$y = 5x - 10$$

x - intercept

y - intercept

$$0 = 5x - 10$$

$$10 = 5x$$

$$x = 2$$

$$\Rightarrow (2, 0)$$

$$y = 5(0) - 10$$

$$y = 0 - 10$$

$$y = -10$$

$$\Rightarrow (0, -10)$$

* هناك طريقتان *

1) يمكن كتابة معادلة Linear على شكلين

إذا عطينا نقطة

$$y - y_0 = m(x - x_0)$$

إذا عطينا slope

$$y = mx + b$$

2) الرسم أي معادلة Linear ← أطلع نقطتين ثم أوجد بينهم

* يفضل للدرجة أطلع التقاطع مع

y-axis x-axis

Example

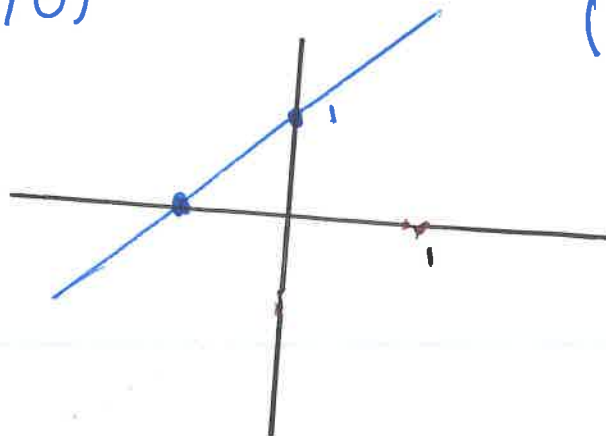
Sketch $y = x + 1$

x-intercept

$$\begin{aligned} 0 &= x + 1 \\ x &= -1 \\ (-1, 0) \end{aligned}$$

y-intercept

$$\begin{aligned} y &= 0 + 1 \\ y &= 1 \\ (0, 1) \end{aligned}$$



(a) Find an equation for the family of linear function with slope 2 and sketch several members of family.

(b) Find an equation for the family of linear function such that $f(2)=1$ and sketch several members of the family.

(c) Which function belongs to ^{both} families?
in part (a) and (b)

Solution

(a) $y = ax + b \Rightarrow y = 2x + b$

if $b=0 \Rightarrow y=2x$

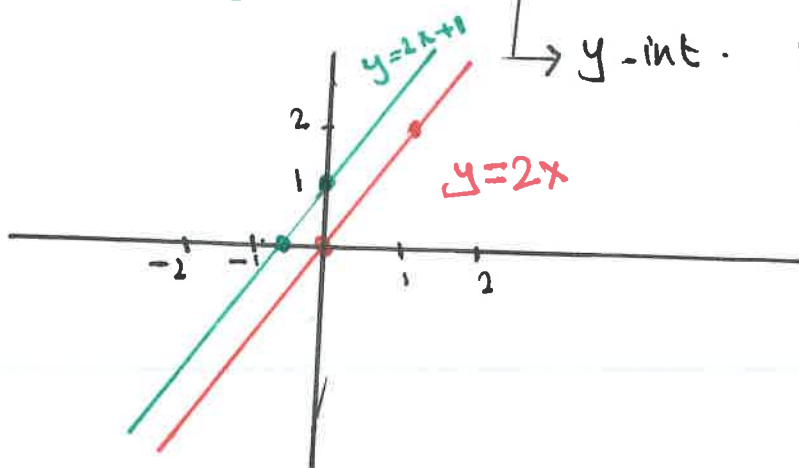
$\xrightarrow{x\text{-int.}} 0=2x \rightarrow x=0$
 $\xrightarrow{y\text{-int.}} y=0 \Rightarrow (0,0)$

$\text{at } x=1 \rightarrow y=2 \Rightarrow (1,2)$

Family of func.

if $b=1 \Rightarrow y=2x+1$

$\xrightarrow{x\text{-int.}} 0=2x+1$
 $x = -\frac{1}{2} \Rightarrow (-\frac{1}{2}, 0)$
 $\xrightarrow{y\text{-int.}} y=0+1$
 $y=1 \Rightarrow (0,1)$



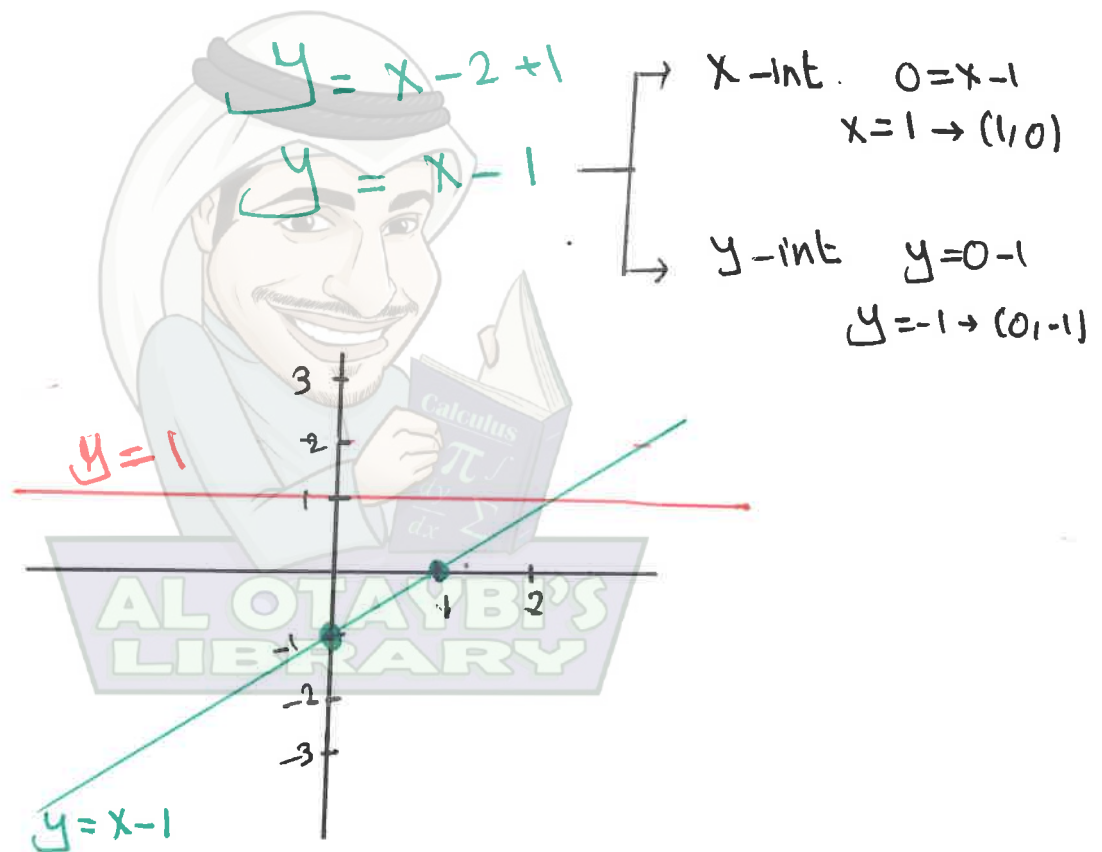
(b) $f(2) = 1 \Rightarrow P(2, 1)$ "معطيات نقطة"

$$y - y_0 = m(x - x_0)$$

$$y - 1 = m(x - 2)$$

if $m = 0 \Rightarrow y - 1 = 0 \Rightarrow y = 1$

if $m = 1 \Rightarrow y - 1 = 1 \cdot (x - 2)$



(c) $m = 2$, $P(2, 1)$

$$y - y_0 = m(x - x_0)$$

$$y - 1 = 2(x - 2)$$

What do all members of the family of linear functions $f(x) = 1 + m(x + 3)$ have in common? Sketch several members of the family.

$$f(x) = 1 + m(x + 3)$$

$\Downarrow$

$$y = 1 + m(x + 3)$$

$$y - 1 = m(x - (-3))$$

$$P(-3, 1)$$

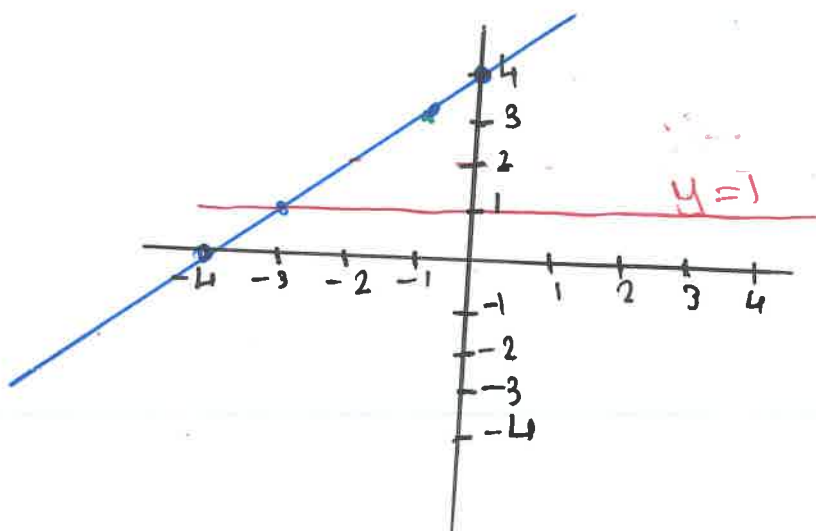
$\Rightarrow$ all members are passing through the point.

if $m = 0 \Rightarrow y = 1 + 0 \Rightarrow y = 1$

if $m = 1 \Rightarrow y = 1 + x + 3$
 $y = 4 + x$

$\rightarrow$ x-int. $0 = 4 + x$
 $x = -4$
 $\Rightarrow (-4, 0)$

$\rightarrow$ y-int. $y = 4 + 0$
 $y = 4$
 $\Rightarrow (0, 4)$



What do all members of the family of linear function $f(x) = c - x$ have in common?

Sketch several members of the family.

$$f(x) = c - x$$

$$y = c - x$$

$$y = -x + c$$

$$m = -1$$

$\Rightarrow$ all members have a slope $= -1$

$$c = 0 \Rightarrow$$

$$y = -x$$

$$\rightarrow \text{x-int. } 0 = -x$$

$$x = 0 \Rightarrow (0, 0)$$

$$\rightarrow \text{y-int. } y = 0$$

$$\text{at } x = 1 \Rightarrow y = -1$$

$$\Rightarrow (1, -1)$$

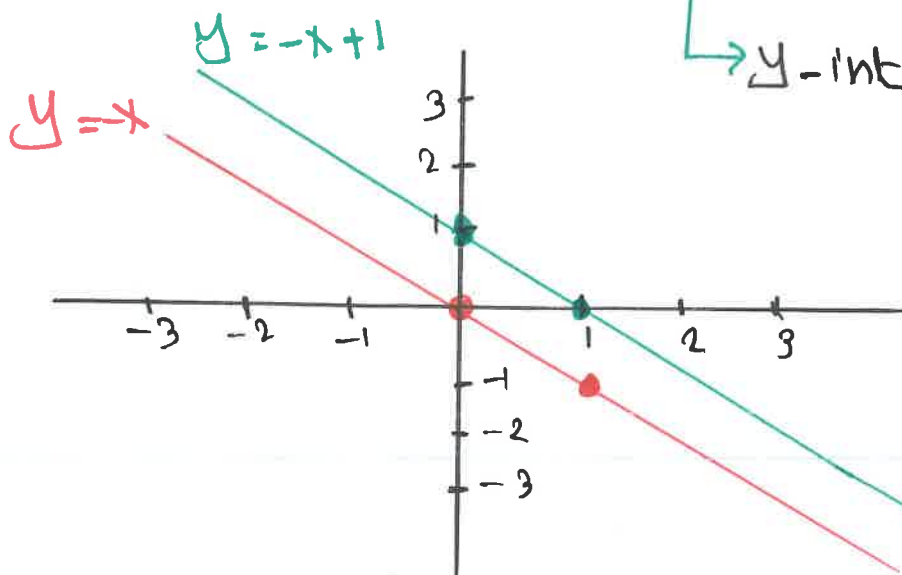
$$c = 1 \Rightarrow y = -x + 1$$

$$\rightarrow \text{x-int. } 0 = -x + 1$$

$$x = 1 \Rightarrow (1, 0)$$

$$\rightarrow \text{y-int. } y = 0 + 1$$

$$y = 1 \Rightarrow (0, 1)$$



Zoom on
Quadratic

* كيفية استخراج المعادلة التربيعية من الرسمية *

معهطيني نقطة الرأس



$$y = ax^2 + bx + c$$

معهطيني نقطة الرأس



$$y = a(x - r)^2 + s$$

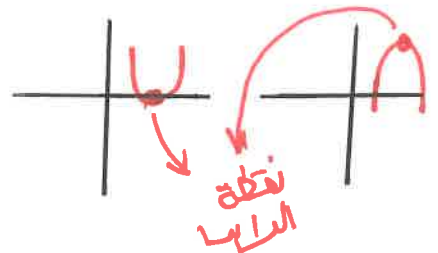
خطواتها:

(1) أعطني الرأس

(2) أعطني

بالنقطة الثانية

$(r, s) \Rightarrow$ نقطة الرأس



Example • Find the expression for the quadratic fun.

$$y = a(x - r)^2 + s$$

$$y = a(x - 3)^2 + 0$$

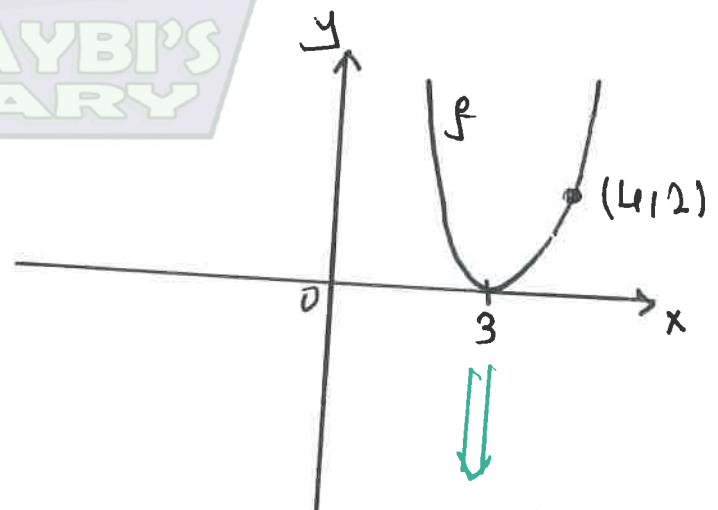
$$y = a(x - 3)^2$$

at P(4,2)

$$2 = a(4 - 3)^2$$

$$2 = a(1)^2 \rightarrow a = 2$$

$$\Rightarrow y = 2(x - 3)^2$$



معهطيني نقطة الرأس

(3,0)

النقطة

(3,0)

(4,2)

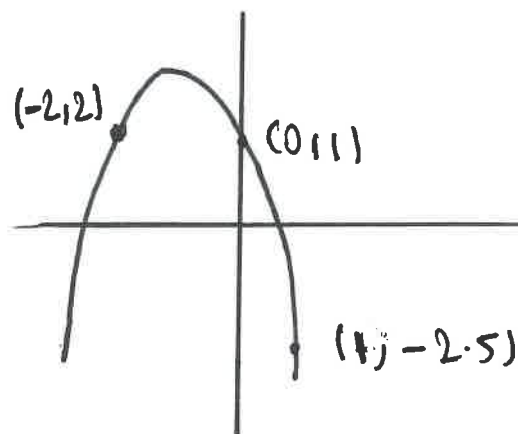
Find the expression for the quadratic function

$$y = ax^2 + bx + c$$

at $P(-2, 2)$

$$2 = a(-2)^2 + b(-2) + c$$

$$2 = 4a - 2b + c \rightarrow \textcircled{1}$$



at $P(0, 1)$

$$1 = 0 + 0 + c \Rightarrow \boxed{c=1} \rightarrow \textcircled{2}$$

↓
نقطة
المعطى

at $P(1, -2.5)$

$$-2.5 = a(1)^2 + b(1) + c$$

$$-2.5 = a + b + 1$$

$(-2, 2)$
 $(0, 1)$
 $(1, -2.5)$

$$a + b = -3.5 \Rightarrow a = -3.5 - b \rightarrow \textcircled{3}$$

② and ③ in ①

$$2 = 4(-3.5 - b) - 2b + 1$$

$$2 = -14 - 4b - 2b + 1$$

$$2 = -13 - 6b \rightarrow 15 = -6b$$

in ③ $\rightarrow \boxed{a = -3.5 - (-2.5) = -1}$

$$b = \frac{-15}{6} \rightarrow b = \frac{-5}{2}$$

$$\boxed{b = -2.5}$$

$$\Rightarrow y = -x^2 - 2.5x + 1$$